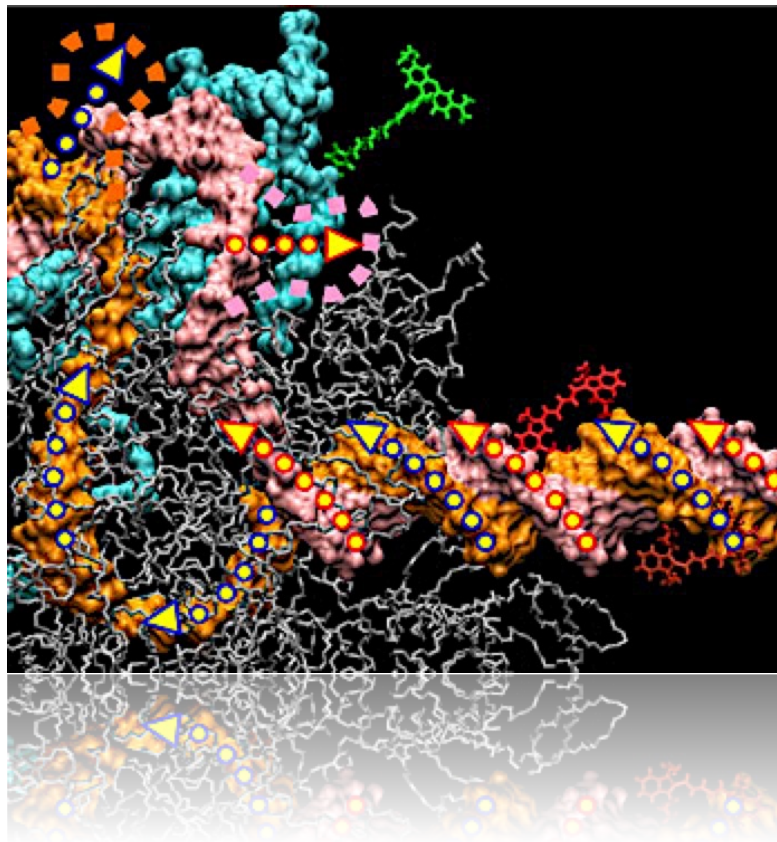


Computational Intelligence II: Evolutionary Optimisation

Chapter 7: Evolutionary Strategies



Overview

- ⌘ This is the oldest type of evolutionary optimisation, especially designed for numerical optimisation of functions.
- ⌘ A simple, good, efficient optimiser, which uses real-valued representation.
- ⌘ It explores the search space using mutation as the only operator, with a population as small as 2 members!
- ⌘ The algorithm starts from an initial point $\mathbf{x}^{(0)} = (x_1, \dots, x_d)$ and proceeds in steps including the following (for minimising F):

- Select a random vector $\mathbf{r}$ from a Gaussian distribution $N(\mathbf{m}, \Sigma)$.
- Set $\mathbf{x}^{(t+1)} = \mathbf{x}^{(t)} + \mathbf{r}$
- If $F[\mathbf{x}^{(t+1)}] < F[\mathbf{x}^{(t)}]$ AND all constraints are satisfied:
- then accept $\mathbf{x}^{(t+1)}$ as the new current point in space,
- otherwise move back as $\mathbf{x}^{(t+1)} = \mathbf{x}^{(t)}$

- ⊙ $\mathbf{m}$ is the mean of the Gaussian (set to 0).
- ⊙ Σ is the covariance matrix, used to set the size of the mutation.
- ⊙ Although this is based on a single member population, it is called a **two-membered ES** or **(1+1)-ES** since the child $\mathbf{x}^{(t+1)}$ competes with its parent $\mathbf{x}^{(t)}$.

Encoding

⌘ Each member $\mathbf{x}$ can be represented as:

$$(\mathbf{x}, \Sigma) = [x_1, \dots, x_d, \sigma_1, \dots, \sigma_k], \quad \text{where } k = d(d+1)/2$$

- ⊙ The covariance Σ includes diagonal (variances) and off-diagonal elements (rotation angles).
- ⊙ By additionally encoding and adjusting Σ , the mutation intensity controlled by Σ , is also co-evolving with the design solution $\mathbf{x}$.
- ⊙ The bigger the elements of Σ , the larger the jumps of the evolutionary search.
- ⊙ In its simplest form (as used in the simple algorithm before), we have:

$$\mathbf{x}^{(t+1)} = \mathbf{x}^{(t)} + N(\mathbf{0}, \Sigma^{(t)})$$

$$\Sigma^{(t+1)} = \Sigma^{(t)} = \sigma \cdot \mathbf{I}_{d \times d} = \text{fixed with equal variances} \\ \text{(and zero covariances)}$$


- ⊙ It can be mathematically proven that:

$$p \left[\lim_{t \rightarrow \infty} F(\mathbf{x}^{(t)}) = F(\mathbf{x}_{\text{optimal}}) \right] = 1$$

- ⊙ Although this is theoretically useful, in practice it is not, since it does not help us to estimate completion times.
- ⊙ Therefore various extensions to help convergence rates and efficiency have been proposed.

Extensions

⌘ 1} The 1/5 success rule ES:

- ◎ It has been observed that the ratio R of successful mutations to all (improving and non improving) mutations should be $1/5$.
- ◎ Therefore, the idea here is to increase the variance of the mutation operator if $R > 1/5$, otherwise decrease it.
- ◎ Thus, the following rule can be applied, with R_k being the success ratio in the last k generations, and $c_{dec} < 1 < c_{inc}$ (with typically $c_{inc} = 1/c_{dec} = 1/0.82 = 1.22$, c_{dec} within $[0.8, 1.0)$):

$$\sigma^{(t+1)} = \begin{cases} c_{dec} \cdot \sigma^{(t)} & \text{if } R_k < 1/5 \\ c_{inc} \cdot \sigma^{(t)} & \text{if } R_k > 1/5 \\ \sigma^{(t)} & \text{if } R_k = 1/5 \end{cases}$$

⌘ 2} ($\mu+\lambda$)-ES:

- ◎ To avoid premature convergence, the enhancement here is a multi-membered architecture, with μ individuals producing λ offspring.
- ◎ The new temporary population of $\mu+\lambda$ members is reduced to μ via a selection process that involves the comparison of their objective values using F .

⌘ Continued...

⌘ 3} (μ, λ)-ES:

- ⊙ This is another multi-membered configuration, where μ individuals produce $\lambda > \mu$ offspring.
- ⊙ The selection process here maintains the best μ from the set of the λ offspring only.
- ⊙ This forces the life of each member to a maximum of one generation. This variation is more suitable to problems with time varying or noisy objective function F , where the optimum may be moving with time.

⌘ 4} ES with crossover:

- ⊙ Crossover can be added to the standard ES, to produce an offspring from two individuals:

$$[\mathbf{x}_1, \Sigma_1] = [X_{11}, \dots, X_{1d}, \sigma_{11}, \dots, \sigma_{1k}]$$

$$[\mathbf{x}_2, \Sigma_2] = [X_{21}, \dots, X_{2d}, \sigma_{21}, \dots, \sigma_{2k}]$$
- ⊙ Then either a discrete GA crossover (e.g., uniform or 2-point), or a real-valued crossover (e.g., intermediate arithmetical) can be applied to these two strings.
- ⊙ Then, we can apply mutation.
- ⊙ But we have to be careful not to invalidate the covariance matrix!

⌘ Continued...

© The entire set of updates of the solution $\mathbf{x}$ and its covariance Σ is:

$$\sigma_{ii}^{(t+1)} = \sigma_{ii}^{(t)} e^{N(0, \Delta\sigma)}$$

$$\alpha_{ij}^{(t+1)} = \alpha_{ij}^{(t)} + N(0, \Delta\alpha), \quad \text{for } i \neq j$$

$$\mathbf{x}^{(t+1)} = \mathbf{x}^{(t)} + N(0, \Sigma^{(t+1)})$$

where:

$$\Sigma = \begin{bmatrix} \sigma_{11} & \dots & \sigma_{1d} \\ \vdots & \ddots & \vdots \\ \sigma_{d1} & \dots & \sigma_{dd} \end{bmatrix}$$

$$\tan(2\alpha_{ij}) = \frac{2\sigma_{ij}}{\sigma_{ii} - \sigma_{jj}}$$

© $\Delta\sigma$ and $\Delta\alpha$ are user-defined step parameters.

© The rotation angles α are used here instead of the covariance off-diagonal elements directly, in order to maintain a positive definite matrix.

<END of Chapter 7>